

Jointly-Layered Iterative Detection and Decoding for Massive MIMO: An EP-SU Based Framework

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Abstract—Expectation propagation with successive updating (EP-SU) offers a symbol-wise, layering-like update, enhancing the detection performance for massive MIMO. Nevertheless, this layered potential has not yet been exploited in iterative detection and decoding (IDD). To address this gap, we propose a jointly-layered IDD algorithm for LDPC-coded massive MIMO systems, termed EP-SU-JL. By restructuring EP-SU in a layered manner and embedding it within a jointly-layered IDD framework, the proposed approach facilitates more frequent and efficient iterative information exchange between the detector and decoder, accelerating convergence and enhancing performance. Additionally, a sorting and grouping scheme is introduced to optimize the symbol update process. Numerical simulations demonstrate that the proposed EP-SU-JL consistently outperforms state-of-the-art (SOA) EP-based IDD algorithms across various MIMO scenarios while maintaining comparable complexity.

Index Terms—Massive MIMO, expectation propagation, EP-SU, iterative detection and decoding, jointly-layered architecture.

I. INTRODUCTION

MASSIVE multiple-input multiple-output (mMIMO) is a key technology in wireless communications, delivering substantial gains in spectral and energy efficiency. However, increasing system scale poses major challenges to detection algorithms, requiring more advanced receiver designs.

Expectation propagation (EP) has emerged as a promising approach for mMIMO detection, utilizing Gaussian approximations via moment matching (MM) to achieve near-optimal performance [1]. Since the introduction of the EP detector (EPD) [1], numerous variants have been developed to enhance detection accuracy and reduce complexity. Improved EP (IEP) [2] and power EP (PEP) [3] refine initialization and MM with minimal additional complexity. Tree-EP [4] applies structured variational inference to improve convergence, whereas EP with approximation (EPA) [5] eliminates matrix inversions while preserving performance. EP with successive updating (EP-SU) [6] leverages the Sherman-Morrison

formula for symbol-wise successive updating, achieving notable performance gains with modest complexity reduction.

Building upon EP detection, iterative detection and decoding (IDD) schemes have been extensively investigated to further enhance receiver performance in channel-coded mMIMO systems [7]. The block EP (BEP) framework introduced a double-loop IDD structure, where decoder feedback is directly incorporated into Gaussian approximations [8]. This was further refined in the double EP (DEP) algorithm, which employs an additional MM step in outer loops to improve posterior information approximation, accelerating convergence [9]. More recently, the EPA-IDD scheme [10] was developed to eliminate matrix inversions during detection iterations and leverage variance convergence properties to enhance performance. However, it still requires an inversion at the beginning of each outer loop. On the other hand, conventional IDD often reaches a performance plateau after a certain number of iterations. Layered decoding schedules have been shown to mitigate this effect by accelerating convergence in specific decoders and IDD schemes [11]. Meanwhile, although EP-SU inherently enables a layering-like symbol-wise update, its potential within the IDD framework remains unexplored.

In this letter, we propose an EP-SU based jointly-layered (EP-SU-JL) IDD algorithm for LDPC-coded mMIMO systems. The principal contributions are as follows: (1) We introduce a jointly-layered IDD architecture and the corresponding layered EP-SU detector, which eliminates matrix inversion operations in the IDD loops. This hierarchical framework enables more frequent information exchange between the detector and decoder during each IDD loop, thereby accelerating convergence. (2) Building upon this architecture, we develop the EP-SU-JL algorithm with a sorting and grouping strategy, in which inter-layer message passing is carefully designed to further enhance performance. (3) Simulation results demonstrate that the proposed EP-SU-JL IDD algorithm consistently outperforms SOA DEP and EPA-IDD algorithms across diverse scenarios in terms of both bit-error-rate (BER) performance and convergence behavior. A complexity analysis is also provided to substantiate the efficiency of the proposed IDD scheme.

II. PRELIMINARIES

A. System Model

Consider an LDPC-coded mMIMO system with N_t transmitting antennas and N_r receiving antennas. At the transmitter, the information bit vector $\mathbf{b}$ is first encoded into a codeword $\mathbf{c}$ using LDPC coding, and then interleaved to obtain $\mathbf{a}$. We denote the code length and rate of $\mathbf{c}$ as L_c and R , respectively.

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The vector $\mathbf{a}$ are grouped into blocks of $Q = \log_2 M$ bits and mapped to transmitted symbols $\tilde{\mathbf{x}} \in \tilde{\mathcal{A}}^{N_t}$ using M -QAM modulation, where $\tilde{\mathcal{A}}$ is the constellation set. These symbols are transmitted over a Rayleigh fading or correlated channel $\tilde{\mathbf{H}} \in \mathbb{C}^{N_r \times N_t}$. Perfect channel state information (CSI) is assumed to be available at the receiver. By utilizing real-valued decomposition, the received signal vector $\mathbf{y} \in \mathbb{R}^{2N_r}$ is obtained by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{H} \in \mathbb{R}^{2N_r \times 2N_t}$, $\mathbf{x} \in \mathcal{A}^{2N_t}$, $\mathcal{A}$ denotes the alphabet for the real and imaginary components of the M -QAM symbols, and $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I}_{2N_r}) \in \mathbb{R}^{2N_r}$ is the additive white Gaussian noise (AWGN). The posterior distribution of $\mathbf{x}$ is given by

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p_r(\mathbf{x}) \propto \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \prod_{i=1}^{2N_t} \mathbb{I}_{x_i \in \mathcal{A}}. \quad (2)$$

where $p_r(\mathbf{x}) \propto \prod_{i=1}^{2N_t} \mathbb{I}_{x_i \in \mathcal{A}}$ is the prior distribution of $\mathbf{x}$, and $\mathbb{I}$ denotes the indicator function.

B. EP and EP-SU MIMO Detection

The essence of EP detection lies in using a tractable Gaussian distribution $q(\mathbf{x})$ to approximate the intractable posterior $p(\mathbf{x}|\mathbf{y})$ through Kullback-Leibler divergence minimization. The approximated joint posterior distribution (AJPoD) $q(\mathbf{x})$ can be factorized as [1]

$$q(\mathbf{x}) \propto \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \exp\left(-\frac{1}{2} \mathbf{x}^T \mathbf{\Lambda} \mathbf{x} + \mathbf{y}^T \mathbf{x}\right), \quad (3)$$

where $\mathbf{y} = [\gamma_1, \dots, \gamma_{2N_t}]^T$ and $\mathbf{\Lambda} = \text{diag}([\lambda_1, \dots, \lambda_{2N_t}])$.

The mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ of $q(\mathbf{x})$ are respectively obtained by

$$\boldsymbol{\Sigma} = \left(\sigma_n^{-2} \mathbf{H}^T \mathbf{H} + \mathbf{\Lambda}\right)^{-1}, \quad (4a)$$

$$\boldsymbol{\mu} = \boldsymbol{\Sigma} \left(\sigma_n^{-2} \mathbf{H}^T \mathbf{y} + \mathbf{y}\right). \quad (4b)$$

To update $(\lambda_i^{(l)}, \gamma_i^{(l)})$ at the l -th iteration, the marginal cavity distribution $q^{(l)}(x_i) = \frac{q^{(l)}(\mathbf{x}_i)}{\exp(\gamma_i^{(l)} x_i - \frac{1}{2} \lambda_i^{(l)} x_i^2)} \sim \mathcal{N}(t_i^{(l)}, h_i^{2(l)})$ is first calculated as

$$h_i^{2(l)} = \frac{1}{1/\Sigma_{i,i}^{(l-1)} - \lambda_i^{(l-1)}}, \quad (5a)$$

$$t_i^{(l)} = h_i^{2(l)} \left(\mu_i^{(l-1)} / \Sigma_{i,i}^{(l-1)} - \gamma_i^{(l-1)} \right). \quad (5b)$$

MM is then performed given the mean $m_i^{(l)}$ and variance $\epsilon_i^{2(l)}$ of $q^{(l)}(x_i) p_r(x_i)$, yielding the updated Gaussian parameters

$$\lambda_i^{(l)} = \beta \left(\frac{1}{\epsilon_i^{2(l)}} - \frac{1}{h_i^{2(l)}} \right) + (1 - \beta) \lambda_i^{(l-1)}, \quad (6a)$$

$$\gamma_i^{(l)} = \beta \left(\frac{m_i^{(l)}}{\epsilon_i^{2(l)}} - \frac{t_i^{(l)}}{h_i^{2(l)}} \right) + (1 - \beta) \gamma_i^{(l-1)}. \quad (6b)$$

where β is the damping factor.

The conventional EP algorithm renews $\boldsymbol{\mu}^{(l)}$ and $\boldsymbol{\Sigma}^{(l)}$ after all pairs of approximated marginal prior distribution (AMPrD) parameters $(\lambda_i^{(l)}, \gamma_i^{(l)})$ are updated. To eliminate the matrix inversion operation in (4a), the EP-SU algorithm [6] applies the *Sherman-Morrison* formula, enabling recursive updates of $\boldsymbol{\mu}^{(l)}$ and $\boldsymbol{\Sigma}^{(l)}$ after each pair $(\lambda_i^{(l)}, \gamma_i^{(l)})$ is renewed. Let $\hat{p}_i^{(l)}(\mathbf{x}|\mathbf{y})$ denote the AJPoD after the update of $(\lambda_i^{(l)}, \gamma_i^{(l)})$ in the l -th iteration. Then:

$$\hat{p}_i^{(l)}(\mathbf{x}|\mathbf{y}) = \hat{p}_{i-1}^{(l)}(\mathbf{x}|\mathbf{y}) \exp\left(-\frac{1}{2} \Delta \lambda_i^{(l)} x_i^2 + \Delta \gamma_i^{(l)} x_i\right), \quad (7)$$

where $\Delta \lambda_i^{(l)} = \lambda_i^{(l)} - \lambda_i^{(l-1)}$, and $\Delta \gamma_i^{(l)} = \gamma_i^{(l)} - \gamma_i^{(l-1)}$.

By (7), the covariance matrix of AJPoD after the update of $(\lambda_i^{(l)}, \gamma_i^{(l)})$, denoted by $\boldsymbol{\Sigma}_i^{(l)}$, can be calculated as

$$\boldsymbol{\Sigma}_i^{(l)} = \left(\left(\boldsymbol{\Sigma}_{i-1}^{(l)} \right)^{-1} + \Delta \lambda_i^{(l)} \mathbf{e}_i \mathbf{e}_i^T \right)^{-1}, \quad (8)$$

where $\mathbf{e}_i$ is the i -th column of the identity matrix. Applying the *Sherman-Morrison* formula to (8), we have

$$\boldsymbol{\Sigma}_i^{(l)} = \boldsymbol{\Sigma}_{i-1}^{(l)} - \frac{\Delta \lambda_i^{(l)}}{1 + \Delta \lambda_i^{(l)} \boldsymbol{\Sigma}_{i-1}^{(l)}(i, i)} \boldsymbol{\Sigma}_{i-1}^{(l)} [i] \boldsymbol{\Sigma}_{i-1}^{(l)} \langle i \rangle, \quad (9)$$

where $[i]$, $\langle i \rangle$, and (i, i) denote the i -th column, i -th row, and (i, i) -th element of a matrix, respectively. After obtaining $\boldsymbol{\Sigma}_i^{(l)}$, the posterior mean of the $(i+1)$ -th transmitted symbol $\boldsymbol{\mu}_i^{(l)}(i+1)$ can be computed as

$$\boldsymbol{\mu}_i^{(l)}(i+1) = \boldsymbol{\Sigma}_i^{(l)} \langle i+1 \rangle \cdot \left(\sigma_n^{-2} \mathbf{H}^T \mathbf{y} + \mathbf{y} \right), \quad (10)$$

where $\mathbf{y}^{(l)} = [\gamma_1^{(l)}, \gamma_2^{(l)}, \dots, \gamma_i^{(l)}, \gamma_{i+1}^{(l-1)}, \dots, \gamma_N^{(l-1)}]^T$.

C. Double-EP Based IDD Algorithms

The DEP IDD algorithm [9] adopts a dual-loop architecture consisting of inner and outer loops. At the t -th IDD iteration, the inner loop employs EPD to iteratively update the AMPrD and generate the extrinsic information $q_E^{(t,L)}(x_i)$ to the decoder, where L is the iteration count of EPD. The outer loop then performs channel decoding and returns the renewed priors $p_r^{(t)}(x_i)$ to the detector. The Gaussian parameters (λ_i, γ_i) are subsequently refined using MM as (6). To reduce complexity, the EPA-based IDD scheme (EPA-IDD) introduced in [10] eliminates the necessity for matrix inversions in the inner loop. By implementing partial resetting and Neumann series approximation (NSA) techniques, EPA-IDD further reduces computational demands while enhancing overall performance.

III. THE PROPOSED EP-SU-JL IDD ALGORITHM

In this section, we first introduce a jointly-layered IDD architecture facilitated by the joint factor graph (JFG). Then a detailed description of the message passing process within the layered EP-SU detector is provided, tailored for this IDD architecture. Finally, by incorporating several enhancement strategies, we present the proposed EP-SU-JL algorithm.

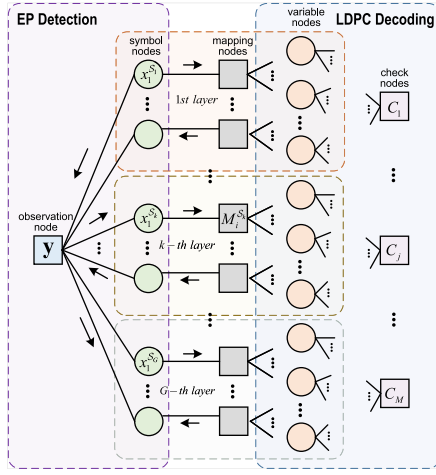


Fig. 1. Joint factor graph of the jointly-layered IDD receiver.

A. Jointly-Layered IDD Architecture

Fig. 1 illustrates the JFG of the jointly-layered IDD receiver, where the observation node (ON) represents the received signal vector $\mathbf{y}$, x_i is defined as the symbol node (SN), and M_i denotes the mapping node (MN). Meanwhile, the variable node (VN) $c_{i,q}$ stands for the q -th demodulated codeword bit of the symbol x_i , and C_j represents the check node (CN).

Let the elements of $\mathbf{x}$ and $\mathbf{c}$ be divided into a total of G sets, and denote $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_G\}$ as a collection of G disjoint index sets. Each element $\mathcal{S}_k$, $k = 1, 2, \dots, G$, encompasses the indices of symbols belonging to group k . All sets are assumed to have equal cardinality S . Unlike the shuffle message passing employed in the original EP-based IDD schemes, the layered IDD algorithm implements a mini-batch information exchange between ON $\mathbf{y}$ and SNs. During each IDD turbo iteration, detection messages are updated and conveyed to the decoder (namely from MNs to VNs) one group at a time. Similarly, only a mini-batch of prior information is fed back from VNs to MNs after LDPC decoding. Therefore, the layered processing of the detector, together with the corresponding layered updating of the decoder inputs and feedback, constitutes the jointly-layered architecture.

B. The Proposed Layered EP-SU Detector

In this subsection, we propose a layered EP-SU detector for the layered IDD scheme. The detailed procedure implemented in the k -th layer of the proposed detector is summarized in Alg. 1. Let the superscript $(\cdot)^{S_k}$ of a vector represent the truncated vector whose elements are selected from the original vector based on the index set $\mathcal{S}_k$. Denote $\Sigma_i^{S_k}$ as the covariance matrix of AJPoD after obtaining the i -th element of λ^{S_k} . Instead of updating only a single AMPrD of a symbol as in (5) and (6) prior to calculating the covariance and mean of AJPoD, the layered EP-SU detector updates $(\mathbf{t}^{S_k(l)}, \mathbf{h}^{2S_k(l)})$ and $(\lambda^{S_k(l)}, \gamma^{S_k(l)})$ in a mini-batch manner as

$$\begin{aligned} \mathbf{h}^{2S_k(l)} &= 1 / \left(1 / \mathbf{V}^{S_k(l-1)} - \lambda^{S_k(l-1)} \right), \\ \mathbf{t}^{S_k(l)} &= \mathbf{h}^{2S_k(l)} \left(\mu^{S_k(l-1)} / \mathbf{V}^{S_k(l-1)} - \gamma^{S_k(l-1)} \right); \end{aligned} \quad (11)$$

Algorithm 1: The Proposed Layered EP-SU Detector (Operations in the k -th Layer)

Input : $\mathbf{y}$, $\mathbf{H}$, σ_n^2 , β , $\mathcal{S}_k$, $\lambda^{(0)} = \mathbf{1}$, $\gamma^{(0)} = \mathbf{0}$, $p_r(x_i)$;

Output: $\mathbf{t}^{S_k(L)}$, $\mathbf{h}^{2S_k(L)}$, $\Sigma_S^{S_k(L)}$;

- 1 **Preprocessing:** Calculate the covariance $\Sigma^{(1)}$ and mean $\mu^{(1)}$ as (4) and initialize $\Sigma_0^{(1)} = \Sigma^{(1)}$;
- 2 **for** $l = 1, 2, \dots, L$ **do**
- 3 Update $(\mathbf{t}^{S_k(l)}, \mathbf{h}^{2S_k(l)})$ in vector form as (11).
- 4 Update $(\lambda^{S_k(l)}, \gamma^{S_k(l)})$ in vector form as (12).
- 5 $\Delta \lambda^{S_k(l)} = \lambda^{S_k(l)} - \lambda^{S_k(l-1)}$.
- 6 **for** $i = 1, 2, \dots, S$ **do**
- 7 Perform (9) to update $\Sigma_i^{S_k(l)}$.
- 8 Calculate the mean vector $\mu^{(l)}$:
- 9 $\mu^{(l)} = \Sigma_S^{S_k(l)} \cdot (\sigma_n^{-2} \mathbf{H}^T \mathbf{y} + \gamma^{(l)})$.

$$\begin{aligned} \lambda^{S_k(l)} &= \beta \left(\frac{1}{\epsilon^{2S_k(l)}} - \frac{1}{\mathbf{h}^{2S_k(l)}} \right) + (1 - \beta) \lambda^{S_k(l-1)}, \\ \gamma^{S_k(l)} &= \beta \left(\frac{\mathbf{m}^{S_k(l)}}{\epsilon^{2S_k(l)}} - \frac{\mathbf{t}^{S_k(l)}}{\mathbf{h}^{2S_k(l)}} \right) + (1 - \beta) \gamma^{S_k(l-1)}, \end{aligned} \quad (12)$$

where $\mathbf{V}^{S_k(l-1)} = \text{diag}(\Sigma_S^{S_k(l-1)})$. Following the parallel computation of AMPrDs whose indices belong to $\mathcal{S}_k$, (9) is applied recursively to update $\Sigma_i^{S_k(l)}$ until all S symbols have been traversed. Finally, each iteration concludes with the update of $\mu^{(l)}$ as in (4b) given $\Sigma_S^{S_k(l)}$.

C. The Proposed EP-SU-JL IDD Algorithm

Leveraging the jointly-layered architecture and the layered EP-SU detector, the proposed EP-SU-JL is summarized in Alg. 2. At the t -th IDD iteration, the message passing process between the k -th layer and the $(k+1)$ -th layer of IDD is detailed as follows:

1) *From the detector to the decoder:* We first select the index set $\mathcal{S}_k$ from $\mathcal{S}$. Let $(\cdot)_i^{S_k}$ denote the i -th element in vector $(\cdot)^{S_k}$. Based on the prior feedback $p_r^{(t)}(\mathbf{x}^{S_k})$ from LDPC decoder, the layered EP-SU detector is performed for L iterations to yield the extrinsic information $q_E^{(t,L)}(x_i^{S_k}) = \mathcal{N}(x_i^{S_k}; t_i^{S_k(t,L)}, h_i^{2S_k(t,L)})$ and the covariance matrix $\Sigma_S^{S_k(t,L)}$. The covariance $\Sigma_S^{S_k(t,L)}$ serves as a global updating parameter and is retained for subsequent layer detection. Specifically, $\Sigma_0^{S_{k+1}(t,0)}$ is initialized by $\Sigma_S^{S_k(t,L)}$, eliminating the need for preprocessing in the layered EP-SU detector (Line 1 in Alg. 1). Notably, the detection parameters at each layer are non-resetting. In the t -th IDD iteration, the Gaussian parameters $(\lambda^{S_k(t,0)}, \gamma^{S_k(t,0)})$ are initialized using the corresponding parameters $(\lambda^{S_k(t-1,L)}, \gamma^{S_k(t-1,L)})$ from the $(t-1)$ -th iteration. Similarly, the posterior mean $\mu^{S_k(t,0)}$ is initialized with $\mu^{S_k(t-1,L)}$. Then the message from SNs to MNs $\phi_{x_i^{S_k} \rightarrow M_i^{S_k}} = q_E^{(t,L)}(x_i^{S_k})$ is converted to LLRs as

$$L_E^{(t)}(c_{i,q}^{S_k}) = \phi_{M_i^{S_k} \rightarrow b_{i,q}^{S_k}} = \ln \frac{\sum \mathcal{X}_q^0 q_E^{(t,L)}(x_i^{S_k})}{\sum \mathcal{X}_q^1 q_E^{(t,L)}(x_i^{S_k})}, \quad (13)$$

Algorithm 2: The Proposed EP-SU-JL IDD Algorithm

Input : $\mathbf{y}, \mathbf{H}, \sigma_n^2, \beta, \lambda^{(0,0)} = 1, \boldsymbol{\gamma}^{(0,0)} = \mathbf{0};$
 $p_r^{(0)}(x_i) = \frac{1}{|\mathcal{A}|} \mathbb{I}_{x_i \in \mathcal{A}}, i = 1, \dots, 2N_t;$

Output: estimated information bit vector $\hat{\mathbf{b}};$

- 1 **Initialization:** Run one DEP iteration to output $\hat{p}^{(1)}(x_i);$
- 2 **for** $t = 1, 2, \dots, T - 1$ **do**
- 3 **Sorting and grouping:** Ascending the variance of the $\hat{p}^{(t)}(x_i)$ and generate $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_G\}.$
- 4 **for** $k = 1, 2, \dots, G$ **do**
- 5 Select the index set $\mathcal{S}_k$ from $\mathcal{S};$
- 6 Run the **layered EP-SU detector** for L iterations given $\mathcal{S}_k$ and $p_r^{(t)}(x_i^{\mathcal{S}_k})$ to output extrinsic information and $\Sigma_{\mathcal{S}_k}^{S_k(t,L)};$
- 7 $q_E^{(t,L)}(x_i^{\mathcal{S}_k}) = \mathcal{N}(x_i^{\mathcal{S}_k}; t_i^{\mathcal{S}_k(t,L)}, h_i^{2\mathcal{S}_k(t,L)});$
- 8 Calculate LLRs $L_E^{(t)}(c_{i,q}^{\mathcal{S}_k})$ as (13);
- 9 Run the **NMS LDPC decoder** for D iterations with $L_E^{(t)}(c_{i,q})$ to obtain $p_r^{(t+1)}(x_i);$
- 10 $\hat{p}^{(t+1)}(x_i) = q_E^{(t,L)}(x_i) p_r^{(t+1)}(x_i);$

where $\mathcal{X}_q^b$ denotes the subset of $\mathcal{A}$ whose q -th bit is $b \in \{0, 1\}$ and $\phi_{M_i^{\mathcal{S}_k} \rightarrow b_{i,q}^{\mathcal{S}_k}}$ is the message passed from MNs to VNs.

2) *From the decoder to the detector:* As $L_E^{(t)}(c_{i,q}^{\mathcal{S}_k})$ is transmitted from the decoder to the detector, the input to the decoder is partially updated. The normalized min-sum (NMS) decoding is then applied for D iterations to derive the prior information $p_r^{(t+1)}(x_i)$. In contrast to the DEP and EPA-IDD schemes, EP-SU-JL eliminates the MM step in the outer loop after decoding, as the prior information provided by the decoder is utilized exclusively by the subsequent detector layer. In other words, only the extrinsic messages from the $(k+1)$ -th layer, denoted by $\phi_{b_{i,q}^{\mathcal{S}_{k+1}} \rightarrow M_i^{\mathcal{S}_{k+1}}} = p_r^{(t+1)}(x_i^{\mathcal{S}_{k+1}}), i = 1, 2, \dots, S$, are required for transmission from VNs to MNs following the k -th IDD layer.

3) *Sorting and grouping:* The order of message updating within the layered sets is crucial, as prioritizing the propagation of more reliable messages typically enhances correctness in subsequent iterations. It is assumed that a smaller variance of the approximated posterior $\hat{p}(x_i)$ indicates a more reliable posterior estimate of x_i . Therefore, at the beginning of each EP-SU-JL IDD iteration, we first sort the variances of $\hat{p}(x_i)$ in ascending order. The corresponding symbol indices are then arranged into an ordered set $\mathcal{S}$, which is subsequently partitioned into G equally-sized subsets in a sequential manner. Notably, to ensure convergence, one DEP iteration is performed as an initialization step to set the detection parameters and produce the initial estimate $\hat{p}^{(1)}(x_i)$.

In our jointly-layered IDD architecture, the decoder serves as a powerful refinement step that interacts with the detector more frequently. This method facilitates the use of strong error-correction information from parity-check constraints to guide the detection process more promptly, enhancing convergence and reducing error propagation, while maintaining overall decoder complexity.

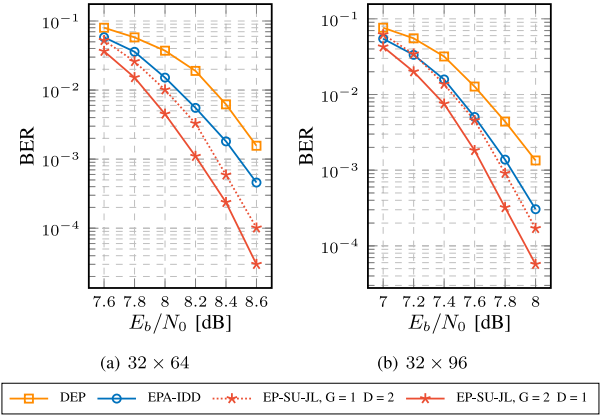


Fig. 2. The BER performance of the proposed EP-SU-JL compared with DEP and EPA-IDD under different antenna configurations with 16-QAM, $L_C=2304$, and $R = 2/3$. The numbers in titles denote $N_t \times N_r$.

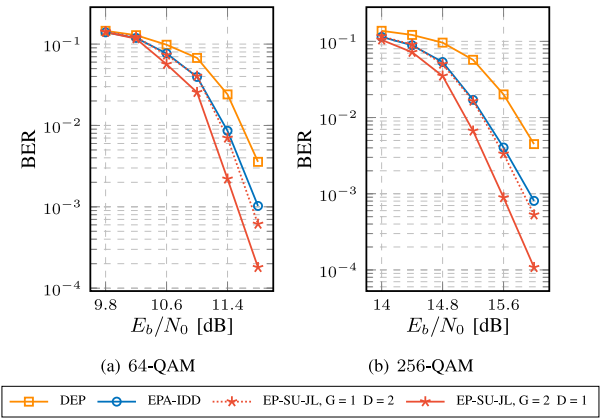


Fig. 3. The BER performance of the proposed EP-SU-JL compared with DEP and EPA-IDD under various modulation orders with $N_t = 32, N_r = 96$, $L_C=2304$, and $R = 2/3$.

IV. SIMULATION RESULTS AND COMPLEXITY ANALYSIS

In this section, we present the simulation results and complexity analysis of the proposed EP-SU-JL IDD algorithm. We compare the BER performance of the EP-SU-JL algorithm with that of DEP and EPA-IDD across various LDPC-coded mMIMO scenarios.

A. Numerical Results

Fig. 2 and Fig. 3 illustrate the BER performance comparisons of different IDD methods in i.i.d. Rayleigh channels. The IDD iteration number T is set to 15 and L is fixed at 1 in all schemes. For DEP and EPA-IDD, we set the decoding iteration per IDD loop D to 2. For EP-SU-JL, both G and D vary while maintaining a product of 2, ensuring the total decoding iterations remain consistent in all methods. In Fig. 2, the simulation is conducted under two different antenna configurations. The proposed EP-SU-JL with $G = 2$ and $D = 1$ outperforms DEP by approximately 0.4 dB and EPA-IDD by about 0.2 dB in both scenarios. Fig. 3 depicts the BER performance of different IDD schemes at higher modulation orders. The proposed EP-SU-JL continues to exhibit superior BER performance compared to DEP and EPA-IDD.

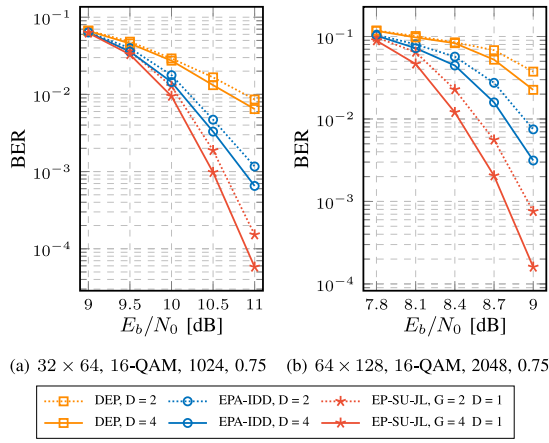


Fig. 4. The BER performance of the proposed EP-SU-JL compared with DEP and EPA-IDD in (a) right-unitarily invariant channel and (b) correlated channel. The numbers in titles denote $N_t \times N_r$, modulation, L_c , and R .

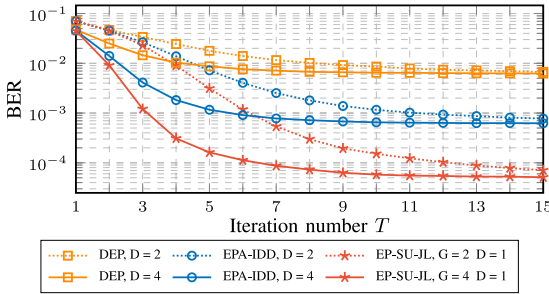


Fig. 5. The BER performance of EP-SU-JL, EPA-IDD, and DEP versus iteration number T at $E_b/N_0 = 11$ dB under the scenario of Fig. 4(a).

Furthermore, in Fig. 4, we explore more general and practical scenarios by considering right-unitarily invariant channels [12] (Fig. 4(a)) and correlated channels [13] (Fig. 4(b)). The correlation coefficients in Fig. 4(b) are set to $\zeta_R = 0.2$ and $\zeta_T = 0.05$. T is fixed at 10 in all cases. Fig. 4(a) demonstrates that EP-SU-JL exhibits a gain of approximately 0.3 to 0.5 dB over EPA-IDD and around 0.9 dB over DEP. In Fig. 4(b), EP-SU-JL with $G = 4, D = 1$ exhibits a gain of 0.35 dB compared to EPA-IDD with $D = 4$. Fig. 5 illustrates the BER performance of DEP, EPA-IDD, and EP-SU-JL versus the number of iterations at $E_b/N_0 = 11$ dB under the scenario of Fig. 4(a). It is clear that EP-SU-JL shows superior convergence behavior, achieving a faster BER reduction than EPA-IDD and DEP for the same total number of decoding iterations.

Notably, as G increases, the signal processing delay rises. Assuming each floating-point operation (FLOP) takes one time unit, and addition, multiplication, or exponentiation counts as one FLOP. The additional time complexity of EP-SU-JL with G layers, compared to a one-layer configuration, is about $\mathcal{O}((\log_2 N_t + Q + 18)(G - 1)T)$ when $L = 1$. Thus, the delay overhead introduced by layering is relatively modest, as the extra delay per layer grows only logarithmically with N_t .

B. Complexity Analysis

Compared to other IDD methods, the decoder in EP-SU-JL exhibits only slight variations in its input interface.

TABLE I
COMPLEXITY COMPARISON OF DIFFERENT IDD ALGORITHMS

Algorithm	Complexity order
DEP	$(2N_t^3 + 2N_t^2 + N_t M)T$
EPA-IDD	$(N_t^3 + 2N_t^2 + N_t M)T$
EP-SU-JL	$N_t^3 + N_t^3(T - 1)/4 + (N_t^2 + N_t M)T$

Consequently, the complexity differences among various methods are primarily observed at the detection stage. The first iteration of EP-SU-JL requires a matrix inversion, which incurs a complexity of $\mathcal{O}(N_t^3)$. For subsequent iterations, the highest complexity order is $\mathcal{O}(N_t^3/4)$ per iteration. Thus, the overall complexity of EP-SU-JL is given by $\mathcal{O}(N_t^3 + N_t^3(T - 1)/4 + (N_t^2 + N_t M)T)$. The overall complexity for different IDD methods when $L = 1$ is listed in Table I. It can be observed that the complexity of EP-SU-JL is lower than that of both DEP and EPA-IDD.

V. CONCLUSION

This letter proposes a novel EP-SU-JL IDD algorithm for LDPC-coded mMIMO systems. Leveraging a jointly-layered architecture and layered EP-SU detector, it eliminates matrix inversions in IDD loops and accelerates iterative information exchange. Simulations and complexity analysis show EP-SU-JL outperforms DEP and EPA-IDD across various mMIMO scenarios with comparable complexity.

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